

NUCLEI

1. Pick out the incorrect statement from the following :
 - (1) β^- emission from the nucleus is always accompanied by a neutrino.
 - (2) The energy of α particle emitted from a given nucleus is characteristic in nature
 - (3) γ ray emission makes the nucleus more stable
 - (4) Nuclear force is charge independent
2. Pick out the incorrect statement from the following :
 - (1) Neutron is the most effective projectile in nuclear reactions.
 - (2) Photon has zero spin, while a neutrino has $\frac{1}{2} \frac{h}{2\pi}$ spin
 - (3) Tritium nuclei do not occur naturally and are prepared artificially in laboratories.
 - (4) A proton can never convert into a neutron as it is lighter than neutron
3. Pick out the incorrect statement from the following
 - (1) Lighter elements are better moderators for nuclear reactors than heavier moderators.
 - (2) In a natural uranium reactor, heavy water is a preferred moderator to ordinary water
 - (3) Mass energy inter-conversion takes place only in nuclear reactions and never in chemical reactions.
 - (4) Cadmium rods are provided in reactors.
4. Pick out the incorrect statement from the following
 - (1) Accurate measurement of atomic masses is carried out by mass spectrometer
 - (2) Energy available from bombardment of beryllium nuclei with α particles is equivalent to that of photons.
 - (3) Isotopes have identical chemical behavior
 - (4) Even we round off mass of neutron and proton to 1 amu, atomic mass of an element may not be an integral multiple of amu
5. With regard to Geiger and Marsden experiment, pick out the incorrect statement from the following :
 - (1) If electrons are used, instead of α particle, size of nuclei can be accurately measured.
 - (2) If α particles with energy much higher than 5.5 MeV are used, significant deviations in calculations may be observed
 - (3) α -particles suffer inelastic scattering against gold nuclei
 - (4) Trajectory of a α -particle depends upon the impact parameter
6. Nuclear forces acting between two nucleons inside nucleus :
 - (1) Has saturation property
 - (2) Is short ranged which explains the constancy of BE/A in the range $30 < A < 170$.
 - (3) Has zero associated potential energy associated when distance between nucleons is 0.8 fm.
 - (4) Is the strongest force inside nucleus
7. For radioactivity, pick the incorrect statement :
 - (1) Radioactivity was discovered by Marie Curie
 - (2) In a spontaneous radioactive decay, mass of products is less than mass of initial nucleus
 - (3) One can never predict which nucleus will undergo decay first
 - (4) ${}_{92}\text{U}^{238}$ is non fissionable but radioactive while ${}_{92}\text{U}^{235}$ is fissionable but non-radioactive
8. With regard to neutrinos, pick the incorrect statement :
 - (1) They are generated during β^+ -decays
 - (2) They have strong interactions with other particles
 - (3) They are neutral particles
 - (4) They are hard to detect
9. In a thermonuclear fusion, which statement is incorrect :
 - (1) Energy is provided by raising the temperature
 - (2) Energy must be provided to overcome coulombic repulsion between approaching positively charged nuclei.
 - (3) Is responsible for the formation of red giant star
 - (4) Required temperature is of the order of 10^8 K.

10. Which of the following statement is incorrect for a nuclear reaction
- (1) If nuclei with less total binding energy transform to nuclei with greater binding energy, there will be a net energy release.
 - (2) Nuclear reactions are not balanced in the same way as chemical reactions
 - (3) Both the number of protons and number of neutrons are conserved, but the total mass is not conserved.
 - (4) In β^- decay a neutrino is generated, while in β^+ decay an anti-neutrino is generated
11. Pick the incorrect statement :
- (1) Fission of 1kg of Uranium generates equal energy as burning of 1kg of coal.
 - (2) During fission of ${}_{92}\text{U}^{235}$, different pairs of intermediate mass fragments can be obtained
 - (3) The peaks at nuclides like ${}_{2}\text{He}^4$, ${}_{8}\text{O}^{16}$ is an evidence of shell like structure inside nuclei.
 - (4) Decay of unstable ${}_{6}\text{C}^{14}$ isotope into stable ${}_{7}\text{N}^{14}$ is the principle of carbon dating.
12. Considering the efficient functioning of a nuclear reactor, which of the following statement is incorrect ?
- (1) Moderators slow down fast neutrons released in a fission reaction via elastic scattering
 - (2) The core of a nuclear reactor is surrounded with reflectors to prevent leakage
 - (3) Heavy water is used as moderators in fast breed reactors that use Plutonium - 239 as fuel
 - (4) Averagely $2\frac{1}{2}$ neutrons are released per fission of a ${}_{92}\text{U}^{235}$ nucleus
13. For the central portion of an atom, called nuclei, pick the incorrect statement
- (1) Nuclei also have discrete energy levels
 - (2) For stability of nuclei, number of neutron: number of proton ratio has to be around 1 : 1 for lighter nuclei and 3 : 2 for heavier nuclei
 - (3) Density of nuclear matter is independent of size of nucleus
 - (4) As mass of proton is smaller than mass of neutron, a proton can never convert into a neutron.
14. Which of the following statement is incorrect for nuclear decays
- (1) γ decay takes place when nucleus in excited state spontaneously decays into ground state
 - (2) After α decay and β decay, γ decay can take place
 - (3) When an electron and positron come together, they can annihilate each other giving energy in the form of γ rays
 - (4) If β^+ emission is energetically allowed, electron capture is necessarily allowed and vice versa.
15. Corresponding to nuclear decays, which of the following statements is incorrect ?
- (1) In α decay, more than 95% of Q-value is realized as kinetic energy of α particle
 - (2) Energy spectrum of α , β and γ particles are discrete line spectrums.
 - (3) In β decay, daughter nucleus is an isobar of parent nuclei.
 - (4) γ rays are electromagnetic radiations of wavelengths shorter than X-rays
16. When a nucleus in an atom undergoes a radioactive decay, the electronic energy levels of the atom
- (1) do not change for any type of radioactivity.
 - (2) change for α and β radioactivity but not for γ radioactivity.
 - (3) change for α radioactivity but not for others.
 - (4) change for β radioactivity but not for others.
17. Heavy stable nucleus has more neutrons than protons. This is because of the fact that
- (1) neutrons are heavier than protons.
 - (2) electrostatic force between protons are repulsive.
 - (3) neutrons decay into protons through beta decay.
 - (4) nuclear forces between neutrons are weaker than that between protons.

18. Suppose we consider a large number of containers each containing initially 50,000 atoms of a radioactive material with a half-life of 1 year. After 1 year,
- (1) all the containers will have 25,000 atoms of the material.
 - (2) all the containers will contain the same number of atoms of the material but that number will only be approximately 25,000.
 - (3) the containers will in general have different numbers of the atoms of the material but their average will be close to 25,000.
 - (4) none of the containers can have more than 25,000 atoms.
19. In a nuclear reactor, moderators slow down the neutrons which come out in a fission process. The moderator used has light nuclei. Heavy nuclei will not serve the purpose because
- (1) they will break up.
 - (2) elastic collision of neutrons with heavy nuclei will not slow them down.
 - (3) the net weight of the reactor would be unbearably high.
 - (4) substances with heavy nuclei do not occur in liquid or gaseous state at room temperature.
20. Outside a nucleus :-
- (1) Neutron is stable
 - (2) Proton & neutron both are stable
 - (3) Neutron is unstable
 - (4) Neither neutron nor proton is stable
21. Fusion reaction takes place at high temperature because :-
- (1) Molecules break up at high temperature
 - (2) Nuclei break up at high temperature
 - (3) Atoms get ionised at high temperature
 - (4) Kinetic energy is high enough to overcome the coulomb repulsion between nuclei
22. Half life of radioactive element depends upon :-
- (1) Amount of element present
 - (2) Temperature
 - (3) Pressure
 - (4) Nature of element
23. In which radioactive disintegration, neutron dissociates into proton & electron :-
- | | |
|-----------------------|------------------------|
| (1) α emission | (2) β^- emission |
| (3) γ emission | (4) Positron emission |